Leicht·Facile by Team Röstigraben Bridgers

Easy-language drafts of Swiss official texts with Apertus, with every fact checked by code

TEAM
Alexander Bobrovsky (bobrovsky@seznam.cz)
Ivan Kuzmych (i.kuzmych@hotmail.com)

REPOSITORY
github.com/bobrovsky420/leicht-facile

DEMO VIDEO
youtu.be/5el7q6Xtgik

MATURITY
Deployable prototype for German and French drafts of public texts and letters. Proof of concept for decisions of child and adult protection authorities (KESB) and courts; these decisions contain strictly confidential personal data (a person's name with details of their family, health and finances) and are a main reason for a sovereign model.

100 % vs 82 %	97 % vs 85 %	0 % vs 15 %	3.7 vs 15.0
fact coverage	**hard facts kept**	**silent losses**	**rule errors**
must-keep facts found	the source's countable values	values lost without a trace	per 100 blocks

Our pipeline compared to a chatbot: the same model with a strong single prompt, on 20 real German texts from Swiss authorities.

1. Summary

Problem. About 1.25 million adults in Switzerland aged 16 to 65 (22%) have low reading skills (Swiss Federal Statistical Office, PIAAC 2022/23); people with intellectual disabilities, language learners and people with dementia need easy language too.

Disability law (Behindertengleichstellungsgesetz BehiG and the UN disability convention) asks authorities for accessible information, and cantons have begun to publish in Leichte Sprache, FALC or lingua facile; each version is written by hand by a few trained writers, who decide which facts the reader needs and must check every deadline, amount and phone number against the original.

Approach. Leicht·Facile is a drafting assistant for the people who write official texts: trained writers, and the case workers, clerks and physicians who write decisions and letters themselves.

One rule: **Apertus does everything that needs language understanding; plain code does everything that can be counted.** Apertus turns the document into a fact sheet that marks the facts the reader must get, writes a draft with provenance for every sentence, judges meaning with a second, larger model, repairs only the flagged sentences and asks again for each value still missing. Code checks the published easy-language rules (45 checks for German, 41 for French), including the exact presence of every fact value. A person reviews source and draft side by side and exports with an audit trail.

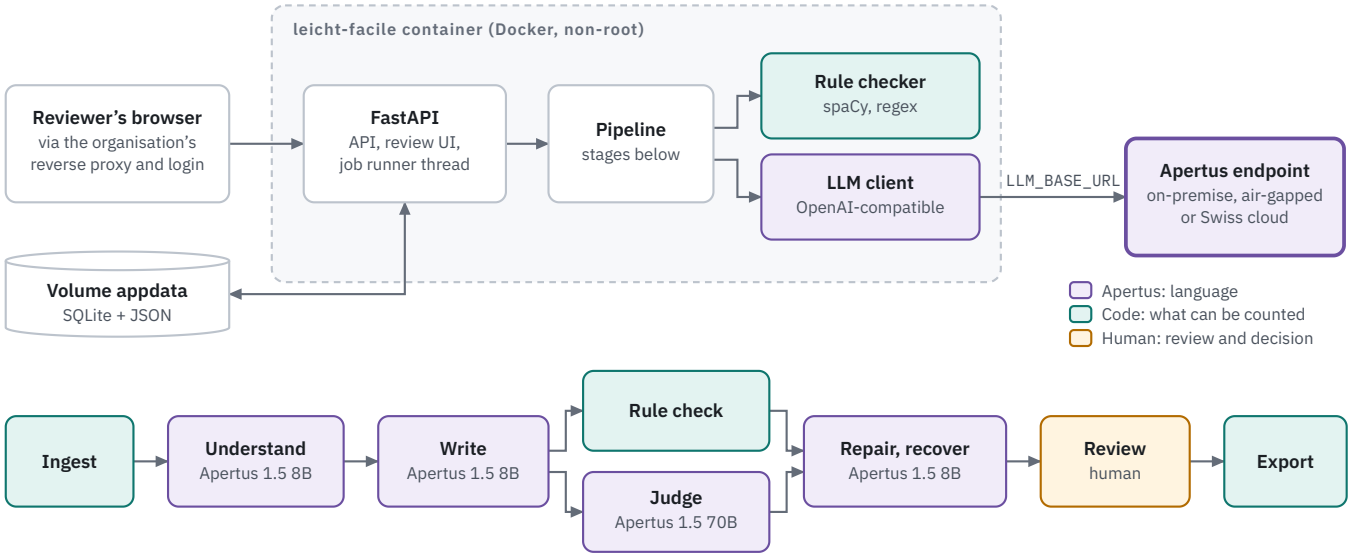
One container plus an OpenAI-compatible Apertus endpoint runs on-premise, air-gapped or on a Swiss cloud, as letters under official, medical or tax secrecy require; for those, only Apertus fits (§3).

The baselines are single prompts on the same model, with no list of the facts that must survive, no provenance and no independent check: a plain one ("rewrite this in Leichte Sprache") and a strong one, the Canton of Zurich's published prompt.

Headline result. Our pipeline compared to the strong single prompt (defined in §5), on the same model and 20 real German texts from Swiss authorities. Our pipeline first, the prompt second:

- **Must-keep facts found:** 100 against 82 percent.
- **Countable values kept** (dates, amounts, phone numbers, addresses): 97 against 85 percent. The official easy-language versions keep 67 percent; they drop values on purpose.
- **Values lost without a trace:** none against 15 percent. The pipeline lists every value it loses for the reviewer.
- **Rule errors per 100 blocks:** 3.7 against 15.0.
- **Invented values per 100 blocks:** 0.0 against 0.0, rare on both sides.
- **Review time:** about 3.5 minutes per 250-word page from opening the draft to the export, in our own timing of three drafts.

2. Architecture



Stage	Done by	Output
Ingest	Code: trafilatura, pdfplumber, spaCy	`SourceDocument`: segments with stable ids (`s0001`)
Understand	Apertus 8B, thinking	`FactSheet`: the reader's core questions, typed facts with normalised values and source ids; code adds every date, amount, contact, address, period or identifier the model missed
Write	Apertus 8B	`Draft`: blocks with the facts and source segments they carry, omissions with reasons, glossary
Rule check	Code: spaCy, regex, wordfreq	Findings with rule id, severity, evidence, suggestion
Judge	Apertus 70B, thinking	Each fact present, missing or distorted; unsupported blocks; meaning findings. The judge never rewrites
Repair	Apertus 8B	New text for flagged blocks only; up to 3 rounds; every version kept
Recover	Apertus 8B	One sentence for each must-keep value still missing, checked and judged like a round
Review, export	Human Code	Per-block decisions; Markdown, HTML, Word and a JSON audit bundle

Key decisions

Meaning in the model, countable rules in code (exact, free, reproducible, no self-grading).

Code checks that every must-keep value is in the draft and that the draft adds none. If the source says "bis spätestens 31. März 2027", the draft must keep that date, as "31. März 2027" or "31.3.2027"; a draft that says "30. März" loses the deadline and adds a date, and code flags both.

Typed artifacts in SQLite let a failed job resume and the UI and the evaluation show every intermediate result.

One service: FastAPI, a server-rendered UI, one worker thread by default; no Node build, no CDN.

Target architecture

Leicht·Facile is deployable in all three architectures. Its only runtime dependency is the Apertus endpoint in `LLM_BASE_URL`; changing the deployment means changing that variable.

Architecture	How it is met	Tested on
a) On-premise	`docker compose --profile local-llm up` adds a vLLM service from the official image `ghcr.io/swiss-ai/vllm_apertus_1.5_release` serving Apertus 1.5 8B on one GPU; the app points at `http://vllm:8000/v1`.	vLLM on one NVIDIA L4 24 GB (rented AWS g6.xlarge in Zurich, Ubuntu 24.04, 10 October 2026): `make run` from a clean clone, two jobs end to end. llama.cpp on a laptop RTX 4060 8 GB.
b) Air-gapped	As (a), with images and weights pulled in advance. Python packages, spaCy models and UI assets are baked into the image; no telemetry, no CDN, no installs at runtime.	Not run without a network. Tests check that the UI loads nothing from another host; CI starts the image against an unreachable endpoint.
c) Sovereign Swiss cloud	`LLM_BASE_URL` points at a Swiss-operated Apertus endpoint.	The CSCS inference API in Lugano: every run evaluated in §5 and the two example letters.

Apart from that endpoint and, for URL input, that page's server, the running system contacts nothing: no telemetry, CDN, outside fonts or scripts, update check or login provider; on-premise or air-gapped, no data leaves the organisation at runtime. Build time: PyPI (`uv.lock`), spaCy wheels from GitHub, images from Docker Hub and GHCR. The container runs without root rights and by default can be reached only from the machine it runs on. On the network it requires a login: a shared token, or the organisation's own login through its reverse proxy.

Licences and providers

Apertus 1.5 Apache 2.0 with an acceptable use policy (accepted once for the gated download), code Apache 2.0, spaCy models MIT and LGPL-LR, fonts OFL. Any OpenAI-compatible server works (used: CSCS, Public AI, llama.cpp; vLLM on-premise): no single provider is needed.

Documentation: architecture, rules, prompts; the full list in the README.

Hardware	Model and server	Memory	Speed
NVIDIA L4 24 GB	8B in bf16, vLLM	18.5 GB of weights; with the FP8 KV cache, one 32k request fits	16 tokens per second, near the limit of the card's memory bandwidth
NVIDIA RTX 4060 8 GB	8B in 4-bit, llama.cpp	5.1 GB of weights	About 37 tokens per second

3. Use of Apertus

- **Models:** `swiss-ai/Apertus-v1.5-8B` writes (Understand, Write, Repair, Recover, baseline) and `swiss-ai/Apertus-v1.5-70B` judges, set by `LLM_NAME` and `LLM_JUDGE_NAME`.
- **How it is used:** inference in a structured-output pipeline; no tool calling, no fine-tuning.
- **Where it runs, and why only Apertus:** CSCS for the hackathon; vLLM, llama.cpp or a Swiss cloud unchanged. Letters that name a person fall under official, medical or tax secrecy (StGB art. 320, 321; DBG art. 110) and may go to a foreign cloud only encrypted with a key the provider lacks (privatim 2025); a model reads them in clear, so the organisation must run it itself and account for what it learned. Only Apertus combines open weights, open training data (website opt-outs honoured), Swiss public origin and German, French and Italian.
- **Why these sizes:** the 8B is enough to read and write. It finds the must-keep facts as well as the 70B does, at half the price per page (§5). The 70B judges, because meaning needs a second, larger model than the one that wrote the draft.

Stage	Model	Thinking	Temperature	Max tokens
Understand	8B	on	0.1	12000
Write	8B	off	0.3	8000
Judge	70B	on	0.0	12000
Repair	8B	off	0.2	6000
Recover	8B	off	0.2	1500
Baseline prompts	8B	off	0.3	6000

Prompts and parsing, in the document's language, each with one filled JSON example instead of the schema. Sentences and blocks carry ids (`s0004`, `b007`): the model cites, code verifies; the judge's unsupported verdicts must quote words that stand in the draft. The writer plans first (the reader's questions as headings) with the level's style guide and two example pairs written by the team; repair gets only the flagged blocks, their findings and facts. Answers are parsed leniently and validated with pydantic (one retry with the error); calls are streamed, retried on broken streams, 429 and 5xx, and logged.

Where Apertus fell short, and what we did about it:

- The 8B drops must-keep facts while writing, ignored compound splits it was given, copied values from the prompt's example and broke JSON: code completes the fact sheet, splits compounds, returns answers with values the document lacks and repairs the JSON; facts rounds and the recover round ask again, a polish round repairs the sentences they add (prompts.md).
- Every fact number comes from code, not from the judge: the 70B judge's fact verdicts agree with blind labels hardly better than chance (kappa 0.08, §5).
- On the local 4-bit build, Write with thinking took 24 minutes and paraphrased; without, 28 seconds and easy language. Write runs without thinking.
- The 8B's drafts are harder to read than the official easy-language versions: their readability score is 52 against 62 (Amstad; higher is easier). With the 70B in every step, the drafts score 57, at twice the price per page.
- On a scanned page sent as one image, both models change digits (in an IBAN or a phone number, for example). Image input is not built (image-input.md).

Other models and coding assistants. No other model at runtime. The 59 labels for the judge's agreement (§5) were made by Claude (Anthropic), blind to the verdicts; no pipeline score depends on them. The code, tests and documents were written with AI coding assistants (Claude Code, Anthropic), directed and reviewed by the team; the assistants also ran the tests and the evaluation scripts; the design, the decisions and every claim here are the team's.

4. Data

Data	Origin	Licence and handling
Reference pairs: **35** real texts with an official easy-language version (**30** German: 20 eval, 10 dev; **5** French: 3 eval, 2 dev)	Cantons Zürich, Bern, St. Gallen, Luzern, Basel-Stadt, Graubünden, Zug, Geneva, Neuchâtel; federal offices BAG, BWL, EBGB	Personal use only. The repository holds only the manifest (`track_2b/data/pairs/manifest.jsonl`: URLs, licence notes); `eval fetch` caches the texts, git-ignored, with a fetch date
Example documents, few-shot examples, hard texts for tests	Written by the team: invented municipality, reserved `.example` domains	In the repository, under the project licence
Recorded runs of the two prepared letters (German, French) and their finished example jobs, so a first run shows a result in seconds	Apertus answers from the CSCS endpoint, 12 October 2026	In the repository; the page says the answers were recorded; an edited letter, other settings or **Run without cache** go to the endpoint

Dev pairs serve prompt development and calibration; eval pairs are held out and used only for the final measurement. No personal data is used. Evaluating against the cantonal and federal web pages, public but under copyright, needs no consent as long as neither the texts nor our rewrites are published (Copyright Act art. 24a and 24d; our reading, not legal advice; sources.md §4).

5. Evaluation

Task. Simplify a standard-language official text to Leichte Sprache and keep every fact. Each held-out pair runs under these conditions:

Code	Condition	Measures
P0	One prompt: “Schreibe diesen Text in Leichter Sprache:”	What a user gets from a chatbot
P0-ZH	The Canton of Zurich’s public Leichte Sprache prompt, verbatim; German only	The strong single prompt: Zurich’s published prompt, the only public Swiss one we found
P1	Understand, Write, rule check; no judge, no repair	Fact sheet, planning, provenance
P2	Full pipeline with judge and repair	Checking and repairing

Two rows need no model: SOURCE (the original, scored as if it were a draft) and REF (the official easy-language version written by professionals).

Metrics, on every row’s final text: **fact coverage** (must-keep facts with a source value found); **hard facts kept** (the source’s countable values, by code, so comparable with REF); **silent losses** (hard values lost without a trace; for the pipeline, the gaps of its code completion); **values not in the source**; **unsupported blocks** (70B judge); **rule errors**; **readability** (Amstad, the German Flesch); **SARI** against the official version (a port of `SARI.py`, checked bit for bit); cost at Public AI’s token prices and speed per 250-word page.

Results on the 20 real held-out German pairs, final run on the CSCS endpoint, 12 October 2026. The strong prompt’s two rows keep the answers of its run of 11 October, because its prompt did not change in between; only the checks ran again on them. Each number is the mean over the pairs; “/100” means per 100 blocks. P1 and the last row reused answers from the cache: P1’s time is not measured, the last row’s is too low.

Setup	Fact coverage	Hard facts kept	Silent losses	Values not in source /100	Unsupported /100 (judge)	Rule errors /100	Amstad	SARI	USD / page	s / page
SOURCE	-	100 %	0 %	0.0	-	31.3	36	-	-	-
REF (official)	-	67 %	33 %	2.8	-	2.7	62	-	-	-
P0	94 %	95 %	5 %	0.2	-	24.9	49	44.3	0.0001	2.8
P0-ZH complete	82 %	85 %	15 %	0.0	-	15.0	47	43.0	0.0003	2.4
P0-ZH complete, 70B	85 %	90 %	10 %	0.3	-	7.0	50	44.7	0.0028	6.0
P0-ZH default	65 %	67 %	33 %	0.0	-	9.0	46	41.5	0.0002	1.4
P1	61 %	59 %	1 %	0.1	-	9.7	50	39.6	0.0012	-
P2 (ours)	**100 %**	**97 %**	**0 %**	**0.0**	**2.7**	**3.7**	**52**	**42.2**	**0.0159**	**59.6**
P2, 70B everywhere	100 %	96 %	0 %	0.0	1.0	0.5	57	42.9	0.0250	60.0
P2, 70B reads, 8B writes	99 %	94 %	0 %	0.0	1.4	3.0	53	42.3	0.0145	31.8

Reading the table:

- The one-sentence prompt and the strong prompt's complete mode keep most values because they rewrite little (8.0 and 7.0 words per sentence; SARI rewards closeness to the source) and lose the rest without a trace; the default mode simplifies more but drops a third of the values, all without a trace.
- Our pipeline keeps more hard facts than the official versions: 97 against 67 percent, and 97 against 61 on the 17 pairs with 3 or more hard facts; the official versions drop values on purpose. It nears their sentence length (7.0 words against 6.1) but is harder to read (Amstad 52 against 62).
- Judge, repair and the recover round (P1 to P2) raise fact coverage from 61 to 100 percent and cut rule errors from 9.7 to 3.7 per 100 blocks; 19 of the 20 final drafts miss no must-keep fact, the twentieth misses 2 of its 23 (one behind the per-round cap of missing facts, one asked for and not written).
- A third of P2's rule errors (13 of 38) come from one two-page health-insurance text; the 7-page source that gave two fifths of the errors on 11 October, rewritten in full into 140 blocks, came out as a 44-block draft with one error this time.
- A single prompt varies from run to run: on unchanged templates, P0-ZH complete found 91 percent of the facts in a first run on 10 October, scored with the checks of that day; the table keeps its answers of 11 October, scored with the final checks: 82 percent (one answer per pair at temperature 0.3).
- French (3 pairs): 100 percent fact coverage, 72 percent of the hard facts (one draft of 117 words for a 489-word source kept 4 of 15, every one listed for the reviewer), 3.5 errors, readability 70 (official 85).

Cost and scale. At Public AI's list prices our pipeline costs 0.0159 USD (about 1.6 US cents) per 250-word page, the recover and polish rounds included; with the 70B everywhere, 0.0250 USD. The cost grows linearly with the number of pages: 10 times the volume costs 10 times as much. On own hardware the GPU is a fixed cost until it is busy. Speed is the limit: at 10 times the volume, jobs wait in a queue, as one runs at a time; running several at once is a setting we have not measured. The writer's time is the cost that matters: our own review of three P2 drafts in the UI (the two example letters and one dev pair, 1,079 source words together), from opening the review to the export, took 15 minutes by stopwatch, about 3.5 minutes per 250-word page (the team, not experts).

Judge reliability. The judge's fact verdicts are a hint for the reviewer, not a measure: every fact number here comes from code, and the review page shows a verdict that code contradicts as a doubt. Its block verdicts, the basis of the "Unsupported" column, agree with blind labels far better. 30 facts and 30 blocks from the P2 dev drafts of 6 October (70B judge), drawn evenly over the verdicts and labelled blind to them (`eval sample`, `eval kappa`). Blocks, supported or not: agreement 80 %, Cohen's kappa 0.60. Facts, present, missing or distorted: 38 %, kappa 0.08 (present or not: 62 %, kappa 0.22).

Checker calibration and tests. On official pages each error is a real deviation or a false positive to fix; calibration cut errors per 100 blocks from 12.6 to 2.0 (30 German pages) and from 5.3 to 2.2 (5 French). 1,454 offline tests include golden sentences for every rule (two or more that fire, one that must not).

6. Limitations

- **Readability with an 8B model.** The drafts keep all must-keep facts and 97 percent of the countable values, list every value they lose, and their sentences are almost as short as the official versions' (7.0 against 6.1 words). What is left is the readability score at level A2: 52 against 62 (Amstad); a larger writer and fine-tuning (§8) are the next steps (20 German eval pairs, CSCS).
- **Not a certification.** Easy-language texts must be tested by readers from the target group; the tool prepares that test, it does not replace it.
- **Italian and B1.** Italian texts get only the language-independent rules (facts, layout, readability). The B1 thresholds are our own; no official Swiss B1 rule set exists.
- **The judge is a model.** Obligations and conditions without a value rely on it alone; many pages carry few countable facts, so the fact-rich subset is reported too.
- **What the numbers rest on.** Fact coverage counts the facts of our own fact sheet, which the baselines never see; hard facts kept, counted by code from the source, is the neutral comparison. The rule checker was calibrated on the official versions, so their low error count is partly by design. The review time comes from three drafts, two of them the example letters, each picked as the cleanest of 30 runs.
- **Decision mode v1.** It is calibrated on one team-written KESB decision and one published court decision; no authority and no reader has seen its output.
- **Not yet supported:** scanned PDFs and photos (§3), an air-gapped image bundle, user accounts and roles.

7. Reproducibility

- **Run:** `LLM_BASE_URL`, `LLM_API_KEY` and `LLM_NAME` in the shell or in `.env`, then `make run`, which opens http://localhost:8000 in the browser once the app is up; `make smoke` runs a bundled letter end to end. `LLM_JUDGE_NAME` is optional: unset, the 70B judges when the endpoint lists it, else the `LLM_NAME` model.
- **Numbers:** `make eval` fetches the pairs, runs P0, P0-ZH, P1 and P2 with the response cache (an interrupted run resumes) and writes the tables; `make calibrate` runs the checker on the official pages without a model.
- **Environment:** Python 3.14 in `python:3.14-slim`, dependencies locked in `uv.lock`, spaCy models pinned by wheel URL; CI runs the linter, the tests and a start of the Docker image on every push.
- **Seeds:** temperatures above zero are not seeded; reruns reproduce exact outputs through the response cache, and each result stores the hash of its prompt template.
- **Hardware and commit:** the evaluation numbers come from the CSCS endpoint, the hardware figures in §2 from our own runs (the L4 run in `track_2b/eval/hardware/`); commit `1f67b11` (the final run of 12 October 2026).

8. Next steps

- **Into use.** The first user we aim for is the cantonal administration of Zurich. Its open-source app simply-simplify-language, by Team Data of the Office for Statistics and Data, rewrites texts through commercial models and allows only public, non-sensitive texts. We are in touch with that team: at their invitation we fixed a bug in the app, and they merged our pull request on 12 October 2026; we have not yet offered them Leicht·Facile. Leicht·Facile adds the fact and rule checks by code and a review with an audit trail. For public texts and standard letters that name no person a cloud endpoint is enough, Public AI included, though it serves part of its Apertus calls from outside Switzerland (architecture §6). Left before the first text goes out: the canton's IT security review and its login proxy in front of the app (the app already reads the user from the proxy's header).
- **Air-gapped installation.** A one-step package; today the images and the weights are moved by hand.
- **Fine-tuning.** A LoRA on the 8B, trained on the drafts that writers approved and the edits they made.
- **The same need elsewhere.** Every canton and federal office that publishes easy language, and texts under secrecy, where only Apertus fits: decisions of KESB and courts (decision mode v1 exists; next an anonymisation step), patient letters, penal and payment orders, social, school and tax decisions. Most depend on a period, a reference number, an address or a dose; code already checks these like a date.
- **Patient letters.** Code already checks doses; left are a drug list, a medical judge (Apertus-70B-MeditronFO, not yet tried) and an assessment under the Medical Devices Ordinance.
- **Italian.** Rules and prompts for lingua facile; Italian texts get only the language-independent checks today.
- **Scanned letters.** Image input: the app cuts each scanned page into horizontal strips small enough that the endpoint keeps every pixel (five per A4 page), and the 8B reads them one by one.

License

Code: Apache License 2.0. This report: Creative Commons Attribution 4.0 (CC-BY-4.0).

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